

A Simple and Detailed Tutorial on Extremum Seeking for Vibrational Stabilization (ES-VS)

From Open-Loop Vibrational Stabilization to Feedback-Controlled Vibrational Stabilization

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Abstract

This supplementary note provides an intuitive tutorial-style explanation of the idea behind extremum seeking for vibrational stabilization (ES-VS), with the goal of making the main mechanism more accessible to readers who may not be experts in nonlinear control theory, averaging, or vibrational systems. We begin with a simple inverted-pendulum example and first recall the classical concept of open-loop vibrational stabilization, where sufficiently fast oscillatory forcing can stabilize an otherwise unstable equilibrium. We then explain why open-loop vibrational stabilization alone is generally not enough if one wishes to stabilize the system around a desired objective. This naturally leads to the ES-VS idea, where vibrational stabilization is combined with a feedback extremum-seeking mechanism. To make the discussion more concrete, we also connect the tutorial viewpoint to the mass-spring-damper and inverted-pendulum examples used in the main ES-VS development, and then briefly link the same idea to the flapping-flight system, where the oscillatory motion arises naturally from the system itself.

1 Purpose of this supplementary note

The purpose of this note is to provide a simple and intuitive explanation of the idea behind *extremum seeking for vibrational stabilization* (ES-VS), which is the main mechanism used in the manuscript. The motivation for preparing this note is that the main paper necessarily presents the theory in a compact and technical way, while some readers, especially those from physics, biology, or related areas, may benefit from a more direct tutorial-style explanation of why ES-VS works and what the corresponding averaged system means.

The central point is the following. Classical open-loop vibrational stabilization shows that oscillatory inputs can stabilize an otherwise unstable configuration. However, open-loop vibrational stabilization alone does not provide a feedback mechanism for *adjusting* or *regulating* that stabilization according to a desired objective. ES-VS adds exactly this missing feedback layer. In this sense, ES-VS can be understood as *vibrational stabilization guided by extremum-seeking feedback*.

To explain this in the simplest possible way, we begin with a pendulum example and use Fig. 1 as a visual guide. In particular, Fig. 1(a) illustrates the natural stability properties of the pendulum, Fig. 1(b) illustrates open-loop vibrational stabilization, and Fig. 1(c) illustrates the feedback-controlled ES-VS idea.

2 A simple inverted-pendulum example

Consider the classical inverted pendulum. As illustrated in Fig. 1(a), under the convention used here the upright position corresponds to $\theta = 0$ and is naturally unstable, whereas the downward

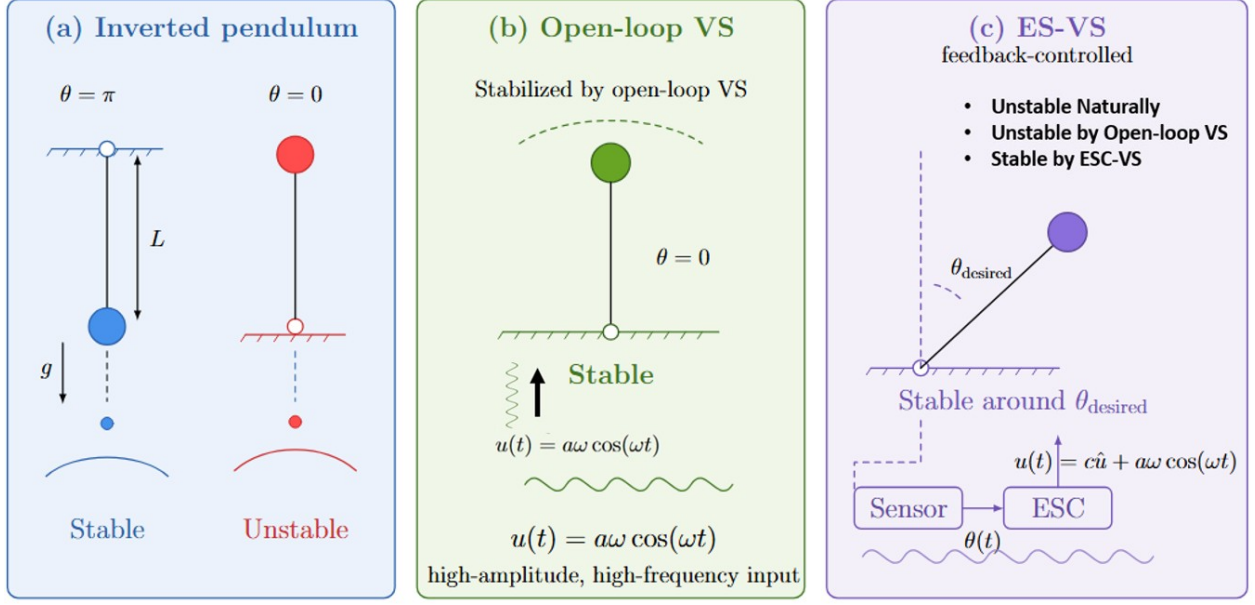


Figure 1: From the natural inverted-pendulum dynamics to open-loop vibrational stabilization and feedback-controlled vibrational stabilization (ES-VS). Panel (a) shows the classical inverted-pendulum convention used here, where the upright equilibrium $\theta = 0$ is naturally unstable and the downward equilibrium $\theta = \pi$ is naturally stable. Panel (b) illustrates open-loop vibrational stabilization, where a high-amplitude, high-frequency input stabilizes the upright position. Panel (c) illustrates ES-VS, where feedback-controlled vibrational stabilization achieves stability around the desired angle θ_{desired} .

position corresponds to $\theta = \pi$ and is naturally stable. This simple fact makes the pendulum an ideal example for understanding the difference between natural stability, open-loop vibrational stabilization, and feedback-controlled vibrational stabilization.

A simplified inverted-pendulum model consistent with the convention used in the main ES-VS development can be written as

$$\ddot{\theta} = \frac{g}{L} \sin \theta - \frac{\beta}{mL} \dot{\theta} + u(t), \quad (1)$$

where θ is the angular position measured from the upright vertical, $\dot{\theta}$ and $\ddot{\theta}$ are the corresponding angular velocity and angular acceleration, g is the acceleration due to gravity, L is the pendulum length, β is the damping coefficient, m is the pendulum mass, and $u(t)$ is an external forcing term.

Without any additional control action, the equilibrium $\theta = 0$ is unstable. The main question is therefore the following:

Can fast oscillatory forcing create an *effective stabilizing mechanism* around an otherwise unstable configuration?

The answer is yes, and this is precisely the main idea of vibrational stabilization [3].

3 Open-loop vibrational stabilization

In classical vibrational stabilization, one applies a high-amplitude, high-frequency input, for example,

$$u(t) = a\omega \cos(\omega t), \quad (2)$$

where a is the oscillation amplitude and ω is the oscillation frequency. This is illustrated schematically in Fig. 1(b), where the vertical oscillatory actuation stabilizes the otherwise unstable upright position.

Even though the actual system motion contains fast oscillations, averaging theory shows that the oscillatory forcing produces an *effective averaged influence* on the slow dynamics. Under suitable conditions, this effective influence can stabilize the upright equilibrium $\theta = 0$. This is the same general idea behind the well-known Kapitza pendulum: fast vibration creates an averaged stabilizing effect that is not obvious from the original unforced system alone [3].

Thus, open-loop vibrational stabilization helps us understand that fast oscillations are not merely disturbances. When properly structured, they can create an effective stabilizing dynamics in the averaged system. For additional visualization of this idea, see the following video: Video demonstration of vibrational stabilization.

4 Why open-loop vibrational stabilization alone is not enough?

Although open-loop vibrational stabilization can stabilize an unstable equilibrium, it does not by itself provide a mechanism for *choosing*, *adjusting*, or *regulating* the stabilization according to a desired target or performance objective.

For example, suppose one does not merely want to stabilize the pendulum near the upright position $\theta = 0$, but instead wants to regulate the motion around a desired angle θ_{desired} , as illustrated in Fig. 1(c), or according to the minimum of some objective function $J(\theta)$. Then pure open-loop oscillation is no longer enough, because the oscillatory forcing in (2) is imposed independently of the measured state and independently of the target objective.

In other words,

- open-loop VS can *create* stabilization,
- but it cannot by itself *adapt* that stabilization using feedback from the system.

This is exactly where feedback becomes essential. If one wants stabilization around a desired configuration such as

$$\theta = \theta_{\text{desired}},$$

then a feedback mechanism is needed to tell the system whether it is moving toward or away from the desired target. Open-loop VS alone does not contain such a mechanism.

5 The main idea of extremum seeking for vibrational stabilization (ES-VS)

The idea of ES-VS is to combine the stabilizing effect of vibrational forcing with the adaptive, feedback-based objective regulation of extremum seeking control (ESC). This is illustrated schematically in Fig. 1(c), where the vibrational input is no longer used purely in open loop, but is shaped by feedback through an ESC-type update law.

In simplified terms, the mechanism works as follows:

1. a high-frequency oscillatory signal creates an effective vibrational stabilization effect;
2. an objective function J is measured from the system state;
3. the control update is driven by J in an extremum-seeking fashion;

- the resulting closed-loop system no longer uses oscillation purely in open loop, but instead uses oscillation together with feedback to regulate the effective stabilized behavior.

A simplified ES-VS structure can be written schematically as

$$u(t) = c \hat{u}(t) + a\omega \cos(\omega t), \quad (3)$$

with the adaptive law

$$\dot{\hat{u}} = k J(\theta) \omega \cos(\omega t), \quad (4)$$

where c , a , and k are design constants.

The interpretation is important:

- the oscillatory term $a\omega \cos(\omega t)$ is what creates the vibrational effect;
- the update law for $\hat{u}$ uses the measured objective J , thereby introducing the missing feedback mechanism.

This is why ES-VS is fundamentally different from open-loop VS:

ES-VS does not only stabilize through vibration; it stabilizes through vibration *shaped by feedback from the objective function*.

6 What the averaged system means?

A key point in understanding ES-VS is that the actual closed-loop system is oscillatory and time-varying, but its slow behavior can be described by an averaged system. In the original ES-VS dynamics, the control input contains a high-amplitude, high-frequency oscillatory term together with a feedback-driven adaptive term. As a result, the trajectory of the full system contains fast oscillations. However, these oscillations do not act merely as rapidly varying disturbances. Instead, when analyzed through averaging, they generate an *effective* contribution to the slow dynamics.

To explain this more clearly, consider the class of second-order mechanical systems used in the main ES-VS development in [1]:

$$\ddot{\mathbf{q}} = \mathbf{f}(\mathbf{q}, \dot{\mathbf{q}}) + \mathbf{u}, \quad (5)$$

where $\mathbf{q} \in \mathbb{R}^n$ is the generalized coordinate vector, $\dot{\mathbf{q}} \in \mathbb{R}^n$ is the generalized velocity vector, $\mathbf{f}(\mathbf{q}, \dot{\mathbf{q}})$ denotes the natural system dynamics, and $\mathbf{u}$ is the control input. In ES-VS, the control is taken in the form

$$\mathbf{u} = \mathbf{C}\hat{u} + \mathbf{A}\omega \cos(\omega t), \quad (6)$$

together with the update law

$$\dot{\hat{u}} = k J(\mathbf{q}) \omega \cos(\omega t), \quad (7)$$

where $\mathbf{C}$ and $\mathbf{A}$ are constant design vectors, $\hat{u}$ is the scalar adaptive state, $k > 0$ is the learning gain, and $J(\mathbf{q})$ is the objective function.

Thus, the ES-VS system consists of two coupled ingredients. The first is the oscillatory forcing term $\mathbf{A}\omega \cos(\omega t)$, which creates the vibrational effect. The second is the adaptive term $\mathbf{C}\hat{u}$, whose evolution is regulated by the objective function through (7). The full system is therefore oscillatory, time-varying, and nonlinear.

Using well-established averaging methods from geometric control theory and variation-of-constants analysis, the corresponding averaged system can be derived; see [1] for the full derivation. Although

we do not reproduce the full derivation here, the resulting averaged system can be written in the form

$$\dot{\bar{\mathbf{x}}} = \underbrace{\begin{bmatrix} \dot{\bar{\mathbf{q}}} \\ \mathbf{f}(\bar{\mathbf{q}}, \dot{\bar{\mathbf{q}}}) + \mathbf{C}\hat{u} \\ 0 \end{bmatrix}}_{\mathbf{z}(\bar{\mathbf{x}})} + \frac{1}{4} \begin{bmatrix} \mathbf{0} \\ \mathbf{M}_{22}\mathbf{A} \\ -2k\nabla_{\bar{\mathbf{q}}}J(\bar{\mathbf{q}})\mathbf{A} \end{bmatrix}, \quad (8)$$

where

$$\bar{\mathbf{x}} = \begin{bmatrix} \bar{\mathbf{q}} \\ \dot{\bar{\mathbf{q}}} \\ \hat{u} \end{bmatrix},$$

and $\mathbf{M}_{22}$ denotes the part of the drift-related Jacobian that appears in the averaging calculation. The bar notation indicates the slow averaged variables.

Equation (8) is important because it shows, in a much simpler way, what the oscillatory ES-VS system is doing on average. In particular, it shows that the fast oscillations generate an additional effective term in the slow dynamics. Therefore, the averaged effect of ES-VS is not simply the original system in (5); rather, it is the original system plus an additional averaged contribution created by the oscillatory input and its interaction with the system dynamics.

This interpretation is very useful. The term

$$\mathbf{f}(\bar{\mathbf{q}}, \dot{\bar{\mathbf{q}}}) + \mathbf{C}\hat{u}$$

may be viewed as the natural mechanical dynamics together with an adaptive force-like contribution. In other words, $\hat{u}$ acts through the vector $\mathbf{C}$ as an additional effective control component that can help regulate the balance of forces in the system. At the same time, the oscillatory perturbation contributes through the averaged term involving $\mathbf{M}_{22}\mathbf{A}$. This is the mathematical expression of the vibrational effect.

The next question is then the following: how is $\hat{u}$ itself regulated? The answer is given by the third component of the averaged system in (8). In particular, the averaged update contains the term

$$-2k\nabla_{\bar{\mathbf{q}}}J(\bar{\mathbf{q}})\mathbf{A},$$

which shows that the adaptation evolves according to the gradient of the objective function. This means that $\hat{u}$ is not arbitrary. Instead, it is adjusted by the objective function in a direction that drives the averaged system toward the desired extremum. In this sense, the feedback law regulates $\hat{u}$ so that the effective force contribution $\mathbf{C}\hat{u}$ becomes exactly what is needed to stabilize the system around the desired configuration.

Therefore, the averaged system explains the ES-VS mechanism in a transparent way:

- the high-frequency oscillation creates an effective vibrational contribution;
- the adaptive state $\hat{u}$ acts as an effective regulating force;
- the update law drives $\hat{u}$ according to the gradient of the objective function;
- together, these effects stabilize the system around the desired target.

For the pendulum viewpoint used in this tutorial, the meaning is especially intuitive. Open-loop vibration can create stabilization of the upright position, as illustrated in Fig. 1(b). But ES-VS goes one step further: through the adaptive state $\hat{u}$, it regulates the effective stabilizing action so

that the system can be stabilized around a desired angle θ_{desired} , as illustrated in Fig. 1(c). Thus, the averaged system is the mathematical object that reveals how oscillation and feedback work together to achieve both stabilization and regulation.

In this sense, the averaged system does not merely simplify the analysis. It also provides the clearest interpretation of why ES-VS works.

7 Illustrative examples: breakdown for the earlier explanations and simulations

To further clarify the idea behind ESC-VS, we now present two representative simulation examples from classical mechanical systems. These examples help visualize the main difference between open-loop vibrational stabilization and feedback-guided vibrational stabilization. In particular, the inverted-pendulum example provides the most direct continuation of the discussion in Fig. 1, while the mass-spring-damper example shows that the same mechanism is not restricted to pendulum-type systems.

7.1 Inverted-pendulum example: meaning of the averaged dynamics and simulation

We begin with the inverted pendulum, which provides one of the simplest and most intuitive illustrations of the ESC-VS idea. As already shown schematically in Fig. 1(a), under the convention used here the upright position corresponds to $\theta = 0$ and is naturally unstable, whereas the downward position corresponds to $\theta = \pi$ and is naturally stable. Figure 1(b) then illustrates the classical open-loop vibrational-stabilization idea, where a sufficiently fast oscillatory input can create an effective stabilizing behavior around the otherwise unstable upright position. However, if one wishes not only to stabilize the pendulum at the upright position, but also to regulate it toward a desired angle θ_{desired} , then feedback becomes essential. This is precisely the role of ESC-VS, as illustrated in Fig. 1(c).

The inverted pendulum can be written as

$$\ddot{\theta} = \frac{g}{L} \sin(\theta) - \frac{\beta}{mL} \dot{\theta} + c_1 \hat{u} + a_1 \omega \cos(\omega t), \quad (9)$$

together with the adaptive update law

$$\dot{\hat{u}} = kJ(\theta)\omega \cos(\omega t), \quad (10)$$

where the objective function is chosen as

$$J(\theta) = (\theta - \theta_{\text{desired}})^2. \quad (11)$$

As explained in the averaged-system discussion, the high-frequency oscillation does not simply act as a rapidly varying signal. Instead, it generates an effective averaged contribution to the slow dynamics. For the present inverted-pendulum example, the corresponding averaged system can be written in the simplified form

$$\ddot{\bar{\theta}} = \frac{g}{L} \sin(\bar{\theta}) - \frac{\beta}{mL} \dot{\bar{\theta}} + c_1 \bar{u}, \quad (12)$$

together with the averaged update law

$$\dot{\bar{u}} = -\frac{1}{2}ka_1(2\bar{\theta} - 4) = -ka_1(\bar{\theta} - 2). \quad (13)$$

Equations (12)–(13) show the same main idea explained in the averaged-system section: the oscillatory ESC-VS input produces an effective slow dynamics in which $\hat{u}$ acts as a regulating force term in the pendulum equation, while the adaptation law drives $\hat{u}$ according to the objective. In other words, $\hat{u}$ is adjusted so that the effective force contribution $c_1\hat{u}$ balances the pendulum dynamics and regulates the motion toward the desired angle. This is exactly the feedback-guided part that is missing in pure open-loop vibrational stabilization.

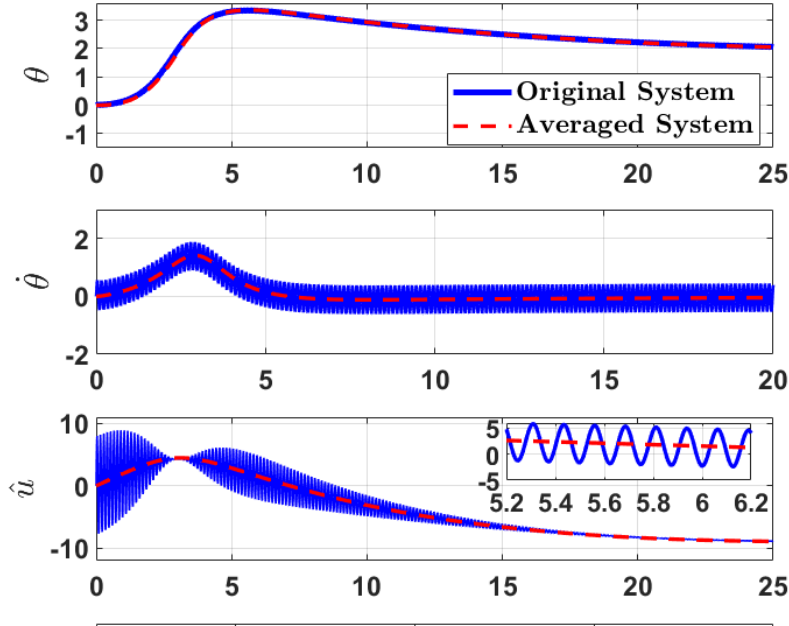


Figure 2: Simulation of the inverted pendulum under ESC-VS. The red dashed curves correspond to the averaged dynamics in (12)–(13). The figure illustrates regulation toward the desired angle θ_{desired} , showing that the oscillatory stabilization effect is adjusted through feedback rather than acting purely in open loop.

Figure 2 shows that the pendulum is not simply kept near the upright position by oscillation alone. Rather, the ESC-VS law uses the objective function to guide the system toward the selected angle θ_{desired} . The red dashed curves in the figure are the trajectories of the averaged model, and they show the slow behavior predicted by the averaged-system interpretation.

7.2 Mass-spring-damper example: meaning of the averaged dynamics and simulation

As a second example, consider the mass-spring-damper system. This example is useful because it shows that the ESC-VS idea is not limited to pendulum-type systems, but applies more broadly to mechanical dynamics. Unlike the inverted-pendulum case, the drift here is already stabilizing by itself. Nevertheless, ESC-VS still plays an important role, because it regulates and shifts the effective stabilization according to the chosen objective.

The system can be written as

$$\ddot{x} = -\frac{\alpha}{m}x - \frac{\beta}{m}\dot{x} + c_1\hat{u} + a_1\omega \cos(\omega t), \quad (14)$$

together with the update law

$$\dot{\hat{u}} = kJ(x)\omega \cos(\omega t), \quad (15)$$

where, for example,

$$J(x) = (x - x_{\text{desired}})^2. \quad (16)$$

Again, following the averaged-system interpretation discussed earlier, the oscillatory forcing produces an effective slow dynamics. For the present mass-spring-damper example, the corresponding averaged system can be written in the simplified form

$$\ddot{\bar{x}} = -\frac{\alpha}{m}\bar{x} - \frac{\beta}{m}\dot{\bar{x}} + \frac{c_1}{m}\bar{\hat{u}}, \quad (17)$$

together with the averaged update law

$$\dot{\hat{u}} = -\frac{1}{2}ka_1(2\bar{x} - 2) = -ka_1(\bar{x} - 1). \quad (18)$$

Equations (17)–(18) have the same interpretation as in the pendulum case. The oscillation creates an effective averaged influence, while the adaptive state $\hat{u}$ acts as a regulating force contribution in the mechanical dynamics. The update law then adjusts $\hat{u}$ according to the objective so that the system is guided toward the desired target. In this way, the oscillation is not used simply to excite the system, but to create, through averaging and feedback, an effective stabilizing and regulating behavior.

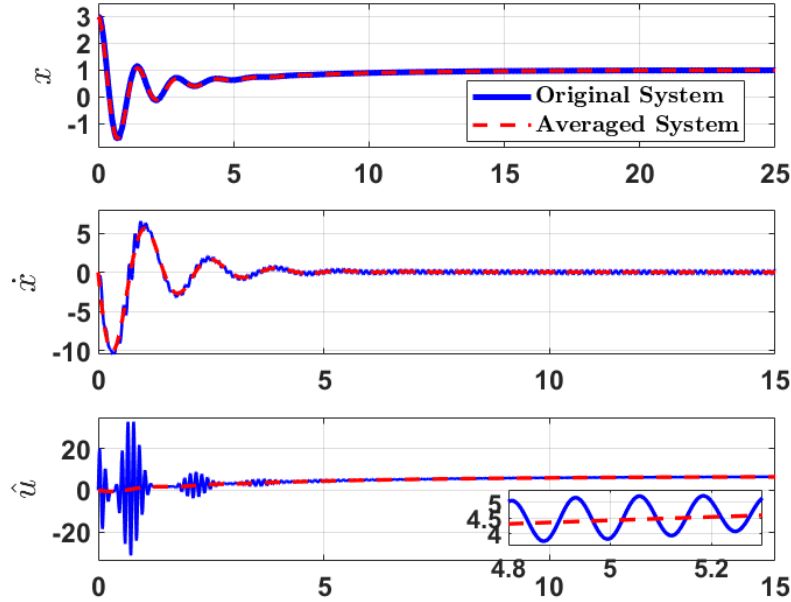


Figure 3: Simulation of the mass-spring-damper system under ESC-VS. The red dashed curves correspond to the averaged dynamics in (17)–(18), illustrating feedback-guided vibrational stabilization toward the desired target.

Figure 3 shows the same central principle observed in the inverted-pendulum example: the averaged stabilizing effect generated by the high-frequency oscillation is shaped by the feedback

update law so that the system evolves toward the desired objective. The red dashed curves in the figure represent the averaged system and therefore show the slow behavior predicted by the analysis. In this sense, the mass-spring-damper example and the inverted-pendulum example illustrate two complementary cases: one in which the drift is already stabilizing but must be redirected toward a desired target, and another in which the drift is destabilizing and the oscillation-based mechanism is essential for stabilization.

7.3 Connection with natural hovering in flapping flight

The same core idea also appears in flapping-flight systems, but with one important difference. In the previous examples, the oscillation was injected artificially through an external oscillatory control term. In the flapping case, however, the oscillation is already naturally present in the system through the wing-flapping motion itself. Thus, there is no need to inject an additional artificial vibration. Instead, the natural flapping dynamics already provide the oscillatory mechanism required for the effective stabilization process, as shown in [2].

Following the flapping model used in [2], the system is described by the altitude z and the wing flapping angle ϕ , and can be written as

$$\begin{bmatrix} \ddot{z} \\ \ddot{\phi} \end{bmatrix} = \begin{bmatrix} -k_{d1}|\dot{\phi}|\dot{z} + g - k_L\dot{\phi}^2 \\ -k_{d3}\dot{z}\dot{\phi} - k_{d2}|\dot{\phi}|\dot{\phi} \end{bmatrix} + \mathbf{u}, \quad (19)$$

where the control input is interpreted through the ES-VS structure. In this setting, the oscillatory mechanism is naturally generated by the flapping motion itself rather than being externally imposed.

This makes the flapping system especially interesting, because it naturally contains the key ingredient required for vibrational stabilization. When combined with an objective-based feedback interpretation, this leads to the natural hovering extremum-seeking viewpoint. In the example shown in Fig. 4, the objective is chosen so that the system is regulated toward the desired altitude $z_{\text{desired}} = 1$, as indicated in the figure.

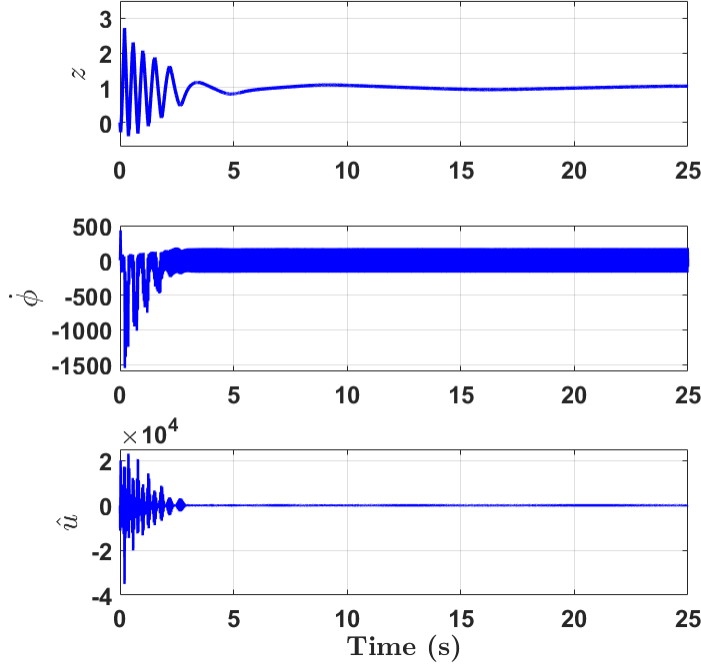


Figure 4: Representative flapping-flight simulation for the source-seeking/hovering case. The system is regulated toward the desired altitude $z_{\text{desired}} = 1$. In this setting, the oscillatory mechanism is naturally present through the flapping motion itself, rather than being injected externally.

Figure 4 illustrates this important point. In contrast to the pendulum and mass-spring-damper examples, where the oscillation is explicitly introduced through the control input, the flapping system already contains an intrinsic oscillatory motion. Moreover, the figure shows that the closed-loop behavior is not merely oscillatory, but is regulated toward the desired altitude $z_{\text{desired}} = 1$. This makes it possible to interpret the resulting stabilization and regulation mechanism through the same ES-VS perspective, but now in a naturally oscillatory system.

8 Concluding Remarks

This tutorial aimed to provide a simple explanation of the main idea behind extremum seeking for vibrational stabilization (ES-VS). Starting from the inverted-pendulum example, we recalled how fast oscillations can create an effective stabilizing influence through vibrational stabilization, and we then explained why open-loop vibration alone is not enough when regulation toward a desired objective is needed.

This naturally leads to the ES-VS idea, where the oscillatory mechanism is combined with extremum-seeking feedback. The averaged system then reveals the effective slow behavior and helps explain why the method can produce both stabilization and regulation.

The examples considered here, including the inverted pendulum, the mass-spring-damper system, and the flapping-flight case, illustrate the same central point: ES-VS can be understood as a feedback-guided vibrational stabilization mechanism.

References

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